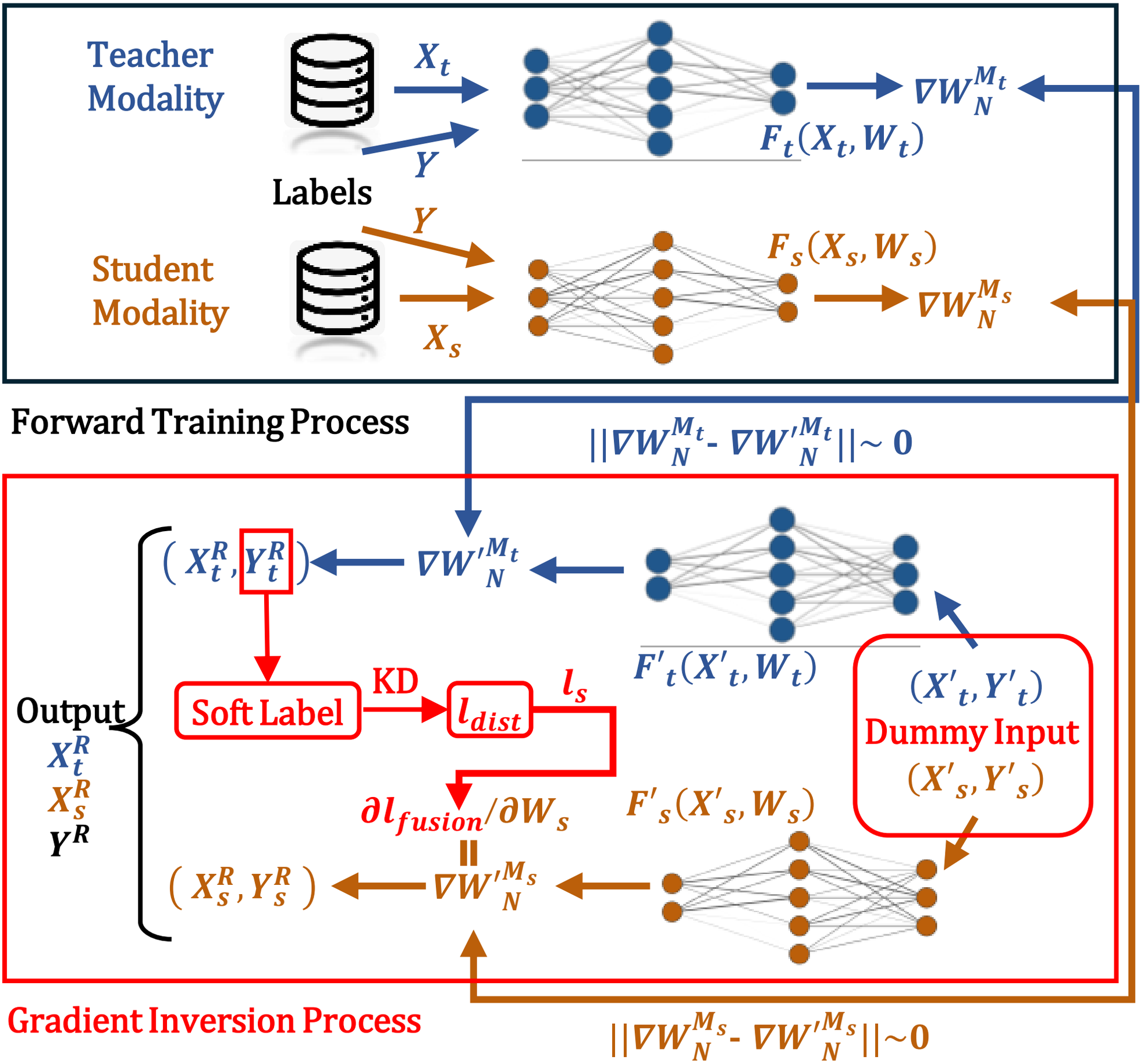

Teacher Modality
X_t
Y
Labels
$F_t(X_t, W_t)$
$\nabla W_N^{M_t}$
Student Modality
Y
X_s
$F_s(X_s, W_s)$
$\nabla W_N^{M_s}$
Forward Training Process
$||\nabla W_N^{M_t} - \nabla W'^{M_t}_N|| \sim 0$
$\nabla W'^{M_t}_N$
(X_t^R, Y_t^R)
$F'_t(X'_t, W_t)$
(X'_t, Y'_t)
Dummy Input
(X'_s, Y'_s)
Soft Label
KD
l_{dist}
l_s
$\partial l_{fusion}/\partial W_s$
$F'_s(X'_s, W_s)$
$\nabla W'^{M_s}_N$
(X_s^R, Y_s^R)
Output
X_t^R
X_s^R
Y^R
Gradient Inversion Process
$||\nabla W_N^{M_s} - \nabla W'^{M_s}_N|| \sim 0$